Phase 8: Data Management & Deployment

🎯 **Goal**: Manage financial transaction data efficiently and move project configurations between environments.

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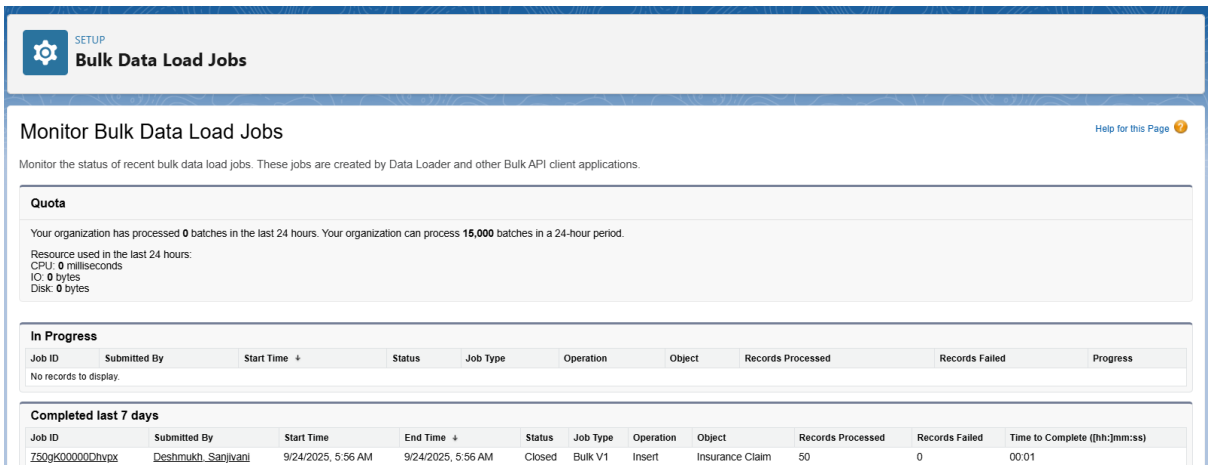
1. Data Import Wizard:

Purpose: Import small demo datasets into Salesforce for testing.

Steps Followed:

- Created a sample CSV with ~50 **demo financial transactions** (Transaction ID, Customer ID, Amount, Location, Risk Score, Fraud Flag).
- Used **Data Import Wizard** (Setup → Data Import Wizard → Custom Objects → Transaction).
- Imported the records successfully (Claim/Transaction ID auto-generated if using Auto Number).

Why Important?
Provides test data for fraud monitoring flows, validations, and dashboards.



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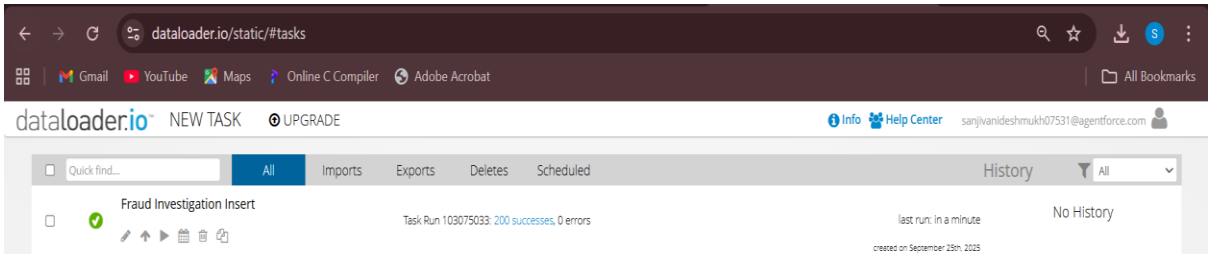
2. Data Loader (Bulk Data):

Purpose: Import or export large datasets (50k+ records).

- Can be used if you want to test system performance with **bulk financial transactions**.
- Supports insert, update, upsert, delete.

Why Important?
Useful for **stress testing fraud detection algorithms** with bigger data.



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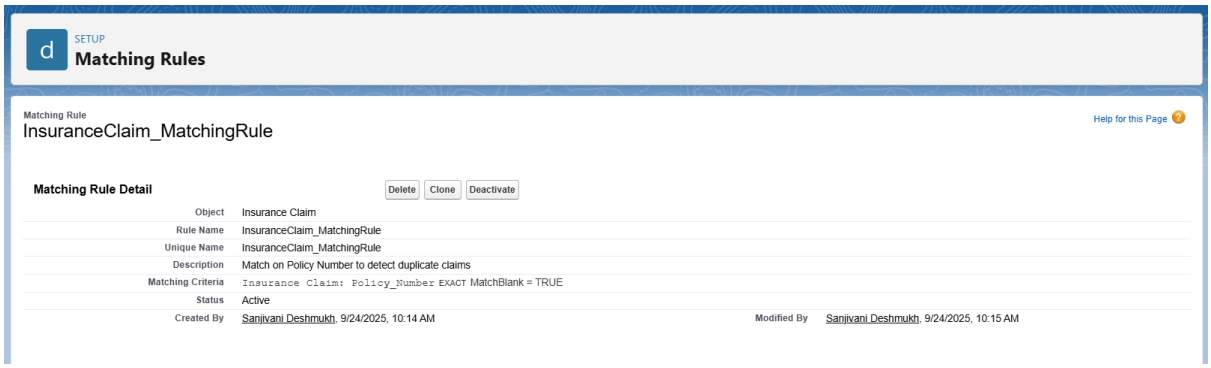
3. Duplicate Rules:

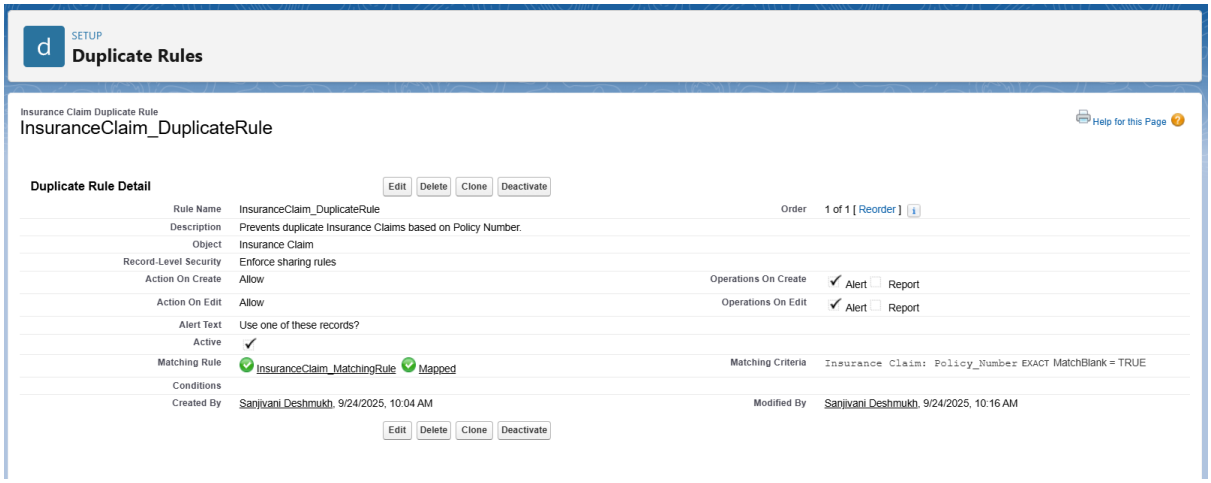
Purpose: Prevent duplicate **transaction records or customer profiles**.

Steps:

- Setup → Duplicate Rules → New Rule.
- For **Customer Profile**, match on Email/Phone.
- For **Transactions**, match on Transaction ID.

Why Important?
Fraud detection depends on accurate data — duplicates could create false alerts.





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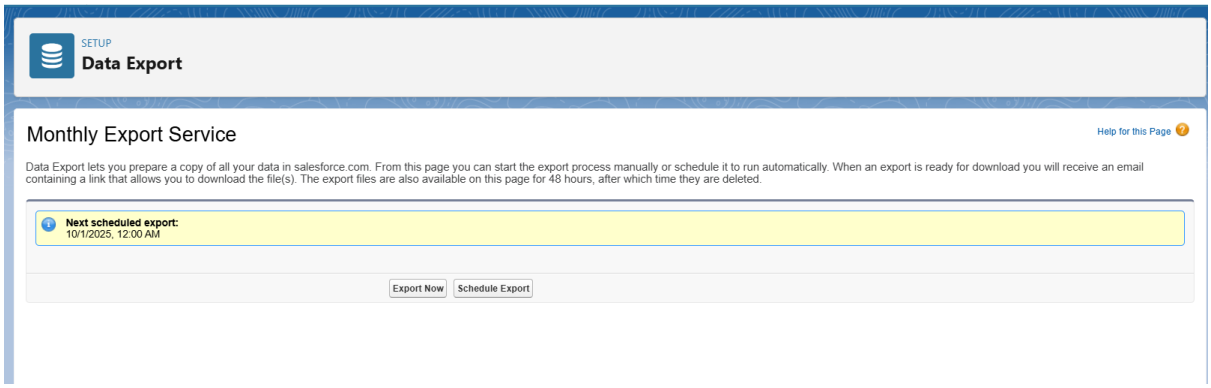
4. Data Export & Backup:

Purpose: Ensure regular backup of fraud data.

Steps:

- Setup → Data Export → Schedule Weekly Export.
- Export all custom objects: Transaction, Customer Profile, Fraud Investigation.
- Store CSV backups securely.

Why Important?
Prevents data loss during errors or deployment, ensures audit history of fraud investigations.



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5. Change Sets:

Purpose:
Change Sets are normally used to move custom objects, fields, Apex classes, flows, and Lightning pages from Sandbox to Production without re-building.

Steps (real-world process):

1. In Sandbox Org → Setup → Outbound Change Sets → New Change Set.
2. Add components: Custom Objects, Apex Classes, Flows, Validation Rules, Approval Processes, Lightning Pages.
3. Upload Change Set to Target Org.
4. In Production Org → Inbound Change Sets → Deploy.

Note for Our Project:
Since we are working in a Developer Edition Org, Change Sets are not available.
Instead, we documented the process for completeness but used VS Code & SFDX to manage deployments in our project.

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6. Unmanaged vs Managed Packages (Optional)

- **Unmanaged:** Share project with classmates → editable.
- **Managed:** Locked → AppExchange style.

Our Case: Use Unmanaged Package if sharing project. No AppExchange publishing needed.

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7. ANT Migration Tool:

Purpose: Command-line deployment tool.

- Used in enterprise teams with CI/CD pipelines.
- Not required for this project.

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8. VS Code & SFDX:

Purpose:
VS Code with Salesforce CLI (SFDX) gives a developer-friendly way to manage Salesforce metadata, automate deployments, and maintain version control with Git. Unlike Change Sets, this approach works in Developer Orgs and is preferred for real projects.

Steps Followed:

1. **Install Salesforce CLI (SFDX):**
    - Download from Salesforce CLI website.
    - Install it on your system.
2. **Install VS Code & Salesforce Extension Pack:**

- Install Visual Studio Code.
- In VS Code → Extensions Marketplace → Search Salesforce Extension Pack → Install.

3. **Authorize Dev Org in VS Code:**

- Open Command Palette in VS Code (Ctrl + Shift + P).
- Type: SFDX: Authorize an Org.
- Choose “Dev Hub” or “Sandbox/Developer Org” → Log in via browser → Allow access.

4. **Create/Connect Project:**

- In VS Code → Command Palette → SFDX: Create Project with Manifest.
- Select a folder for your project.
- This generates a manifest/package.xml file.

5. **Pull Metadata from Org:**

- In the terminal (inside your VS Code project folder), run:

sf project retrieve start --manifest manifest/package.xml

- This fetches all metadata defined in package.xml (e.g., custom objects, Apex classes, LWCs).